

Powder Metallurgy:

Some time it is not possible to achieve the production of desired properties or characteristics by any other manufacturing process like casting, forging etc then the powder metallurgy process is used to achieve the desired properties in the manufactured products.

Powder metallurgy is defined as the process of converting fine metallic powders into useful desired products which involves the mixing of powders of various metals in appropriate proportions, then compacting these mixtures into desired shape in a die under the application of pressure, these compacted parts contained in the die are then heated in a furnace to bind the particles together to impart strength and other properties to the product. During heating the temperature is kept below melting temperature (70 to 80% of the melting point). The products made from this process are very costly because of the high cost of metal powders, technique used for producing powders and as well as the die used. The application of this process is economical only when the number of products required are in bulk i.e. more than ten thousand.

Some of the typical examples of products manufactured by using this process are Tungsten Carbide cutting tools, oilless self lubricating bearings, Turbine blades having high temperature strength and special electrical contacts.

For manufacturing of parts through powder metallurgy it can be classified into two main groups.

- 1) Primary Processes.
2. Secondary Processes.

The primary processes of powder metallurgy are listed below:

- (1) Production of metal powders.
- (2) Mixing or blending of metal powders in required proportions.
- (3) Pressing and compacting the powders in desired shape into a die. It is also known as briquetting.
- (4) Sintering the compacted parts in die to the temp. below melting point under non oxidizing atmosphere. Proceeding the sintered products to secondary processing if required.

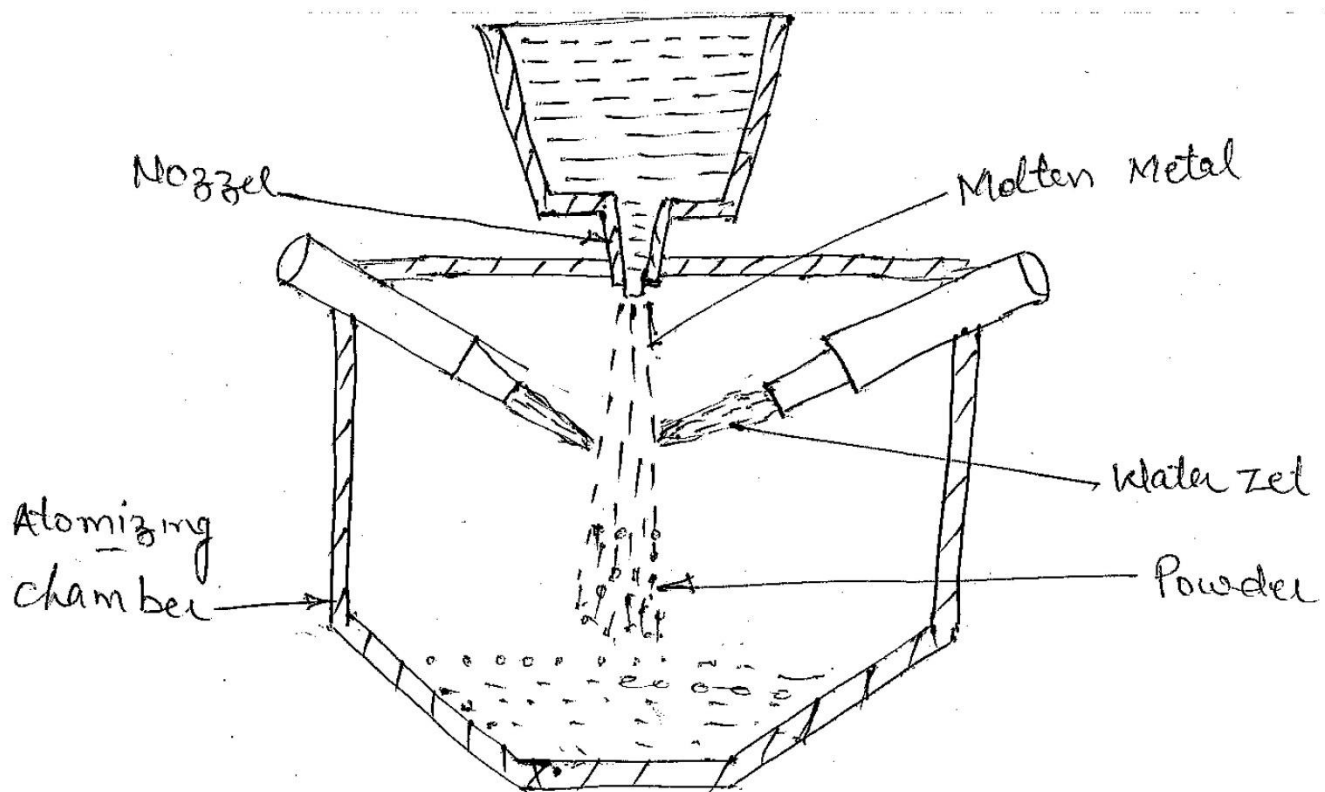
The Secondary processes of powder metallurgy are listed below

- | | | |
|-----------------|-----------------|-------------------|
| 1. Sizing | 2. Coining | 3. Machining |
| 4. Infiltration | 5. Impregnation | 6. Heat treatment |
| 7. Plating | | |

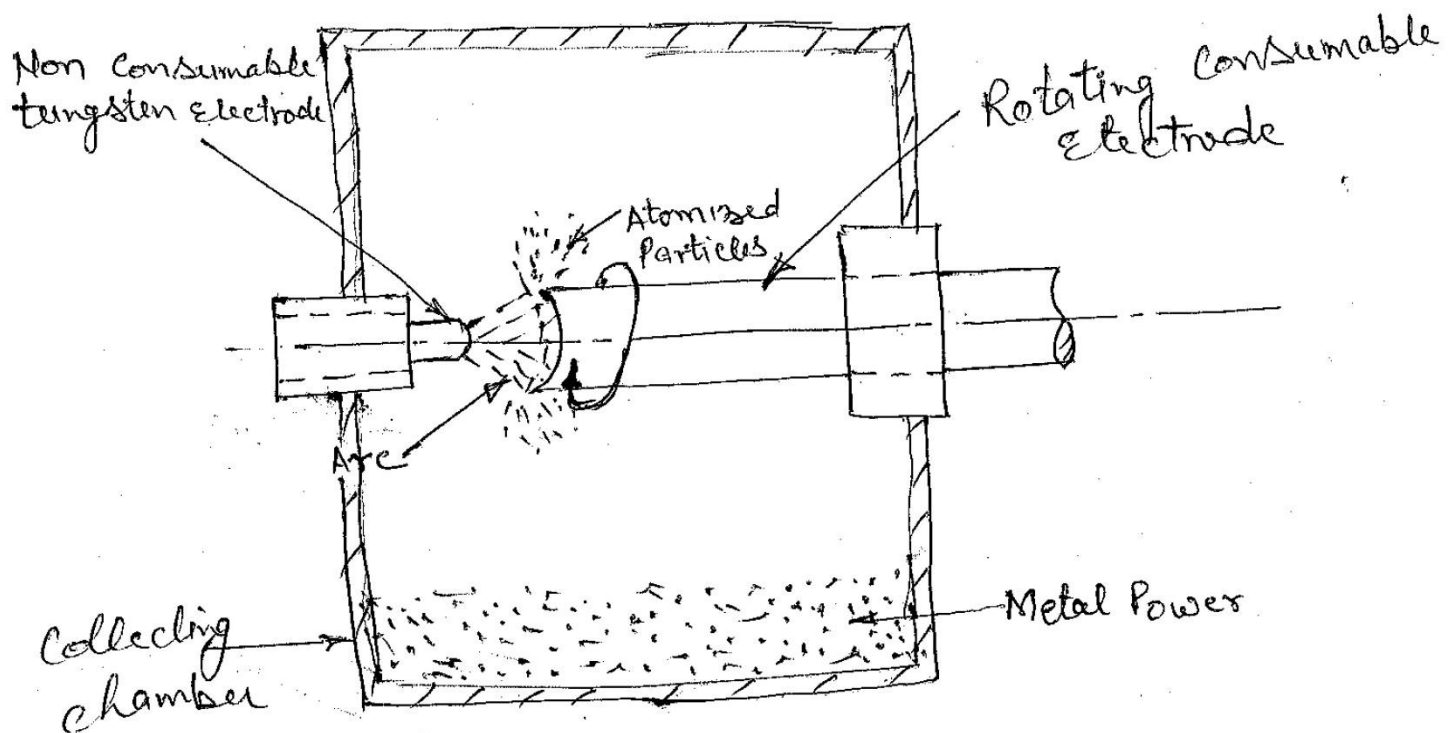
Production of Metal Powders:

Different techniques of producing metal powders are

(1) Atomization: It consists of flowing the molten metal through a nozzle into the stream of air or water as shown in fig. Molten metal is filled in a hopper like container called fundish, fitted on the container top. Metal flows down in the form of stream through the nozzle, water strikes from sides on the metal stream causing rapid cooling in the form of small particles, which are collected in the atomizing chamber. The shape & size of particles depends upon pressure & velocity of water rate of flow of metal, size of nozzle. The shape of particles in this process is generally spherical.



② Rotating Electrode Process (Melt Atomization); It consists of two electrodes one is of tungsten non consumable and the other is made up of metal whose atomization is to be done. The tungsten electrode is stationary whereas the other electrode is consumable and rotating at a very fast speed of about 15000 rpm. An arc is struck between these two electrodes due to which intense heat is produced. The whole set up is enclosed into a chamber provided with inert gas. The metal melts from the rotating electrode which is cooled and changed into small particles and collected at the bottom of the chamber.



Rotating electrode process of powder production

Reduction Process: It is a chemical process used for the production of metal powders in which refined ores or metal oxides are heated with a gas at a high temperature which is below the melting point of metal to be reduced. The gas reacts with metal oxides and reduces the oxygen hence metal becomes free of oxides.

These oxide free metals may be converted into powdered form having uniform fine particle size by the application of mechanical processes like crushing or grinding or pulverizing etc.

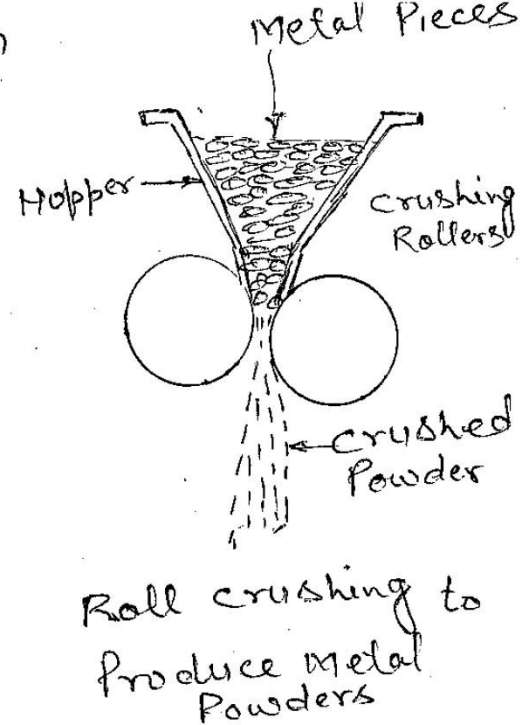
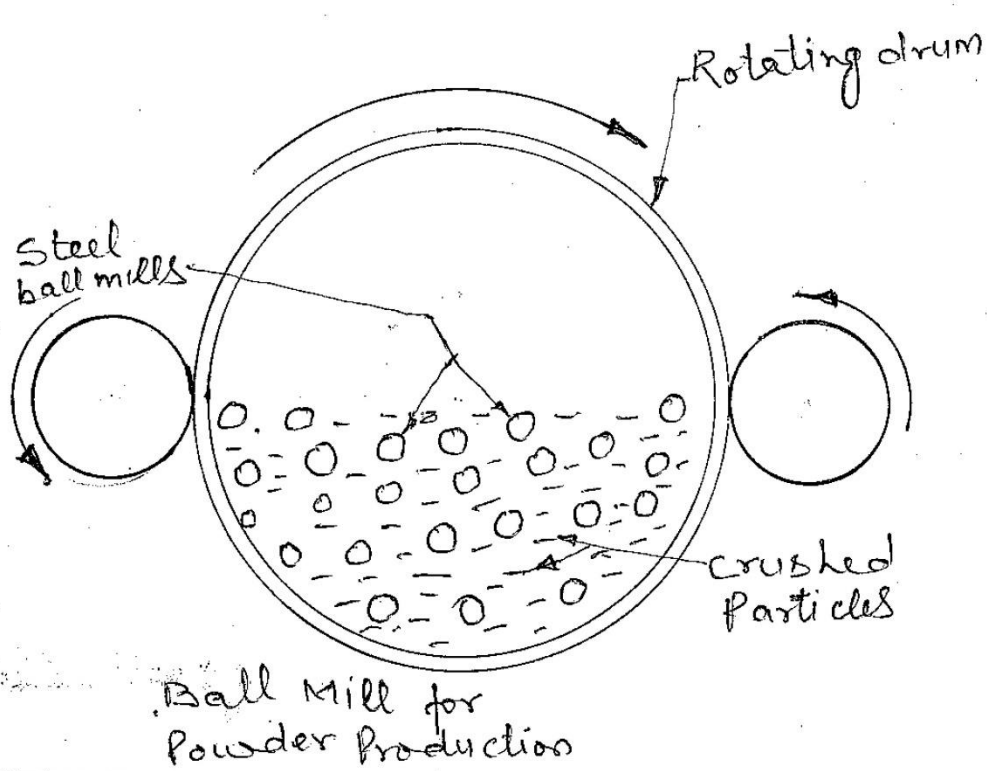
The common reducing process is reduction of oxide compounds by using hydrogen gas.

Metals like tungsten, cobalt, molybdenum, nickel and iron are produced to powder form by the application of this process.

Mechanical Process or Milling and grinding Process:
In this process metals are firstly converted into small particles by crushing milling or rolling processes.

These crushed particles so obtained are filled in a closed cylindrical drum along with steel ball mills and rotated with a very high speed. During the rotating operation these crushed particles are disintegrated in fine particles by the impact of balls. As the process pulverise the crushed particle mechanically hence it is also known as mechanical pulverisation. This process is mainly used for brittle metals. however it can be used for ductile and malleable metals. The particles of size 0.001 mm can be produced by this process

Fig on next page.



Electrolysis process. It is used for the production of extremely pure powders of copper and iron etc. Electrolysis is similar to electroplating. For making copper powder, copper plates are placed as anodes in a tank of electrolyte where as aluminium plates are placed into electrolyte to act as cathodes.

High amperage produces a powdery-deposit of anode metal at cathodes. After a definite time period the cathode are taken out from the tank, are rinsed to remove electrolyte and are dried. The copper deposited on the cathode plates is then scrapped off and pulverized to produce powder of desired grain size.

Shotling: Shotling is the process of dropping the molten metal through a fine sieve into water. A number of metals are reduced to powder form by the application of this process but size of particles obtained is generally large. These particles are further subjected to another mechanical means to produce required fine particles.

Blending or Mixing: It is the process of mixing the powders of different metals in required proportion.

Blending become necessary when either the powders of different metals having same particle size or the same metal for different particle sizes are to be mixed. Blending imparts uniformity in the shape of products. It may be dry or wet. In wet mixing some binders or lubricants are added to meet the following requirements.

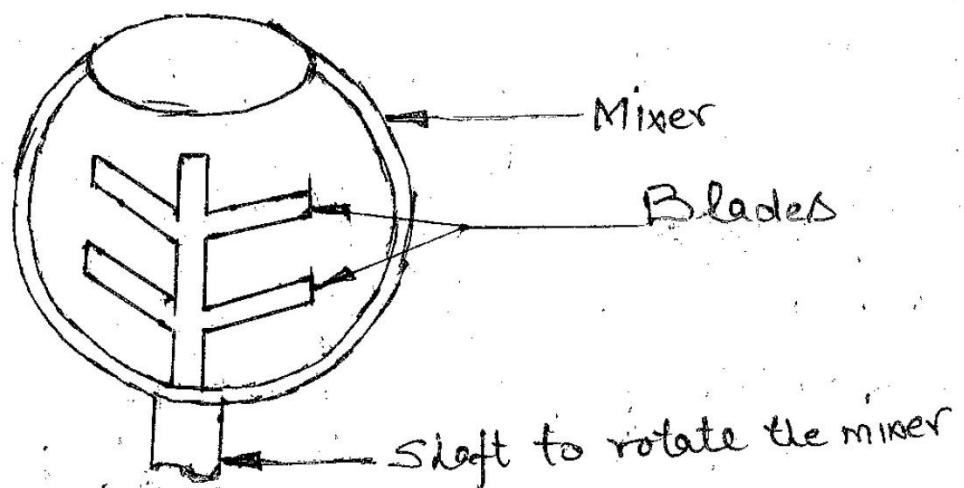
Lubricants: To improve the flow characteristics of powder particles and to reduce the friction between particles and die.

Binders: To enhance the green strength during compaction process.

Several mixing devices used in blending are listed below

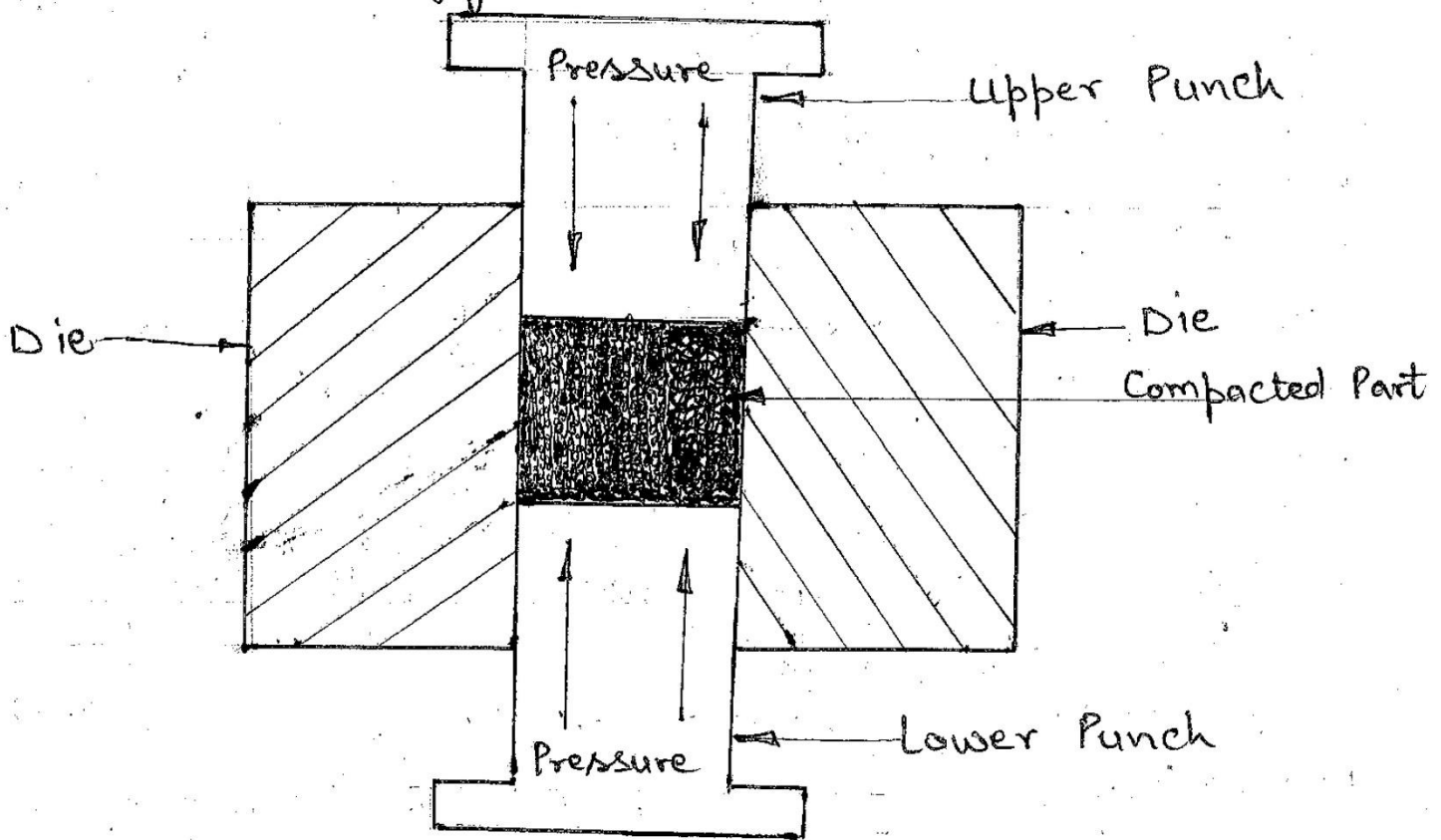
1. Rotary drum
2. Rotating double cone mixer
3. Screw mixer
4. Blade mixer

The blade mixer is shown in fig below.



Compacting or Briquetting or Pressing: It is the process by which the blended powders are pressed in the dies (of required shape) under high pressure with the help of punches to form the desired shape. The process may

consist either single punch or two punches for compaction. The two punches compaction operation is shown in the fig



The punches and dies should be well polished to provide good surface finish & ejection purpose.

The volume between dies should be kept 3 to 5 times than the volume of final compacted product.

The punches are operated either by mechanical presses or by hydraulic operated presses.

Minimum clearance is kept between dies and punches to maintain proper alignment of the product.

- **Pre-Sintering:** It is the process of heating the compacted part to a temperature below the sintering temperature, the objective of this process is to remove the excess lubricants & binders added during blending and to increase green strength of compacted part. If machining of the product is not required then this

operation can be avoided. It is mainly required for selected metals or like tungsten carbide. Tungsten carbide can be easily machined in pre-sintering state because the machining of such metals after sintering become difficult due to excellent hardness after sintering.

Sintering: It is the process by which a compacted part is heated up to a suitable temperature which is below the melting point of powder having higher melting point but higher enough to allow the bonding of particles, in large continuous furnaces under protective atmosphere to avoid oxidation & chemical reactions.

Generally sintering temperature varies from 70% to 90% of the melting point of the powder having high melting point. During sintering process a number of metallurgical changes occur in the compacted part. As a result of atomic diffusion this process metallurgical bond develop among the powder particles which increases the mechanical, electrical and physical properties of the compacted part.

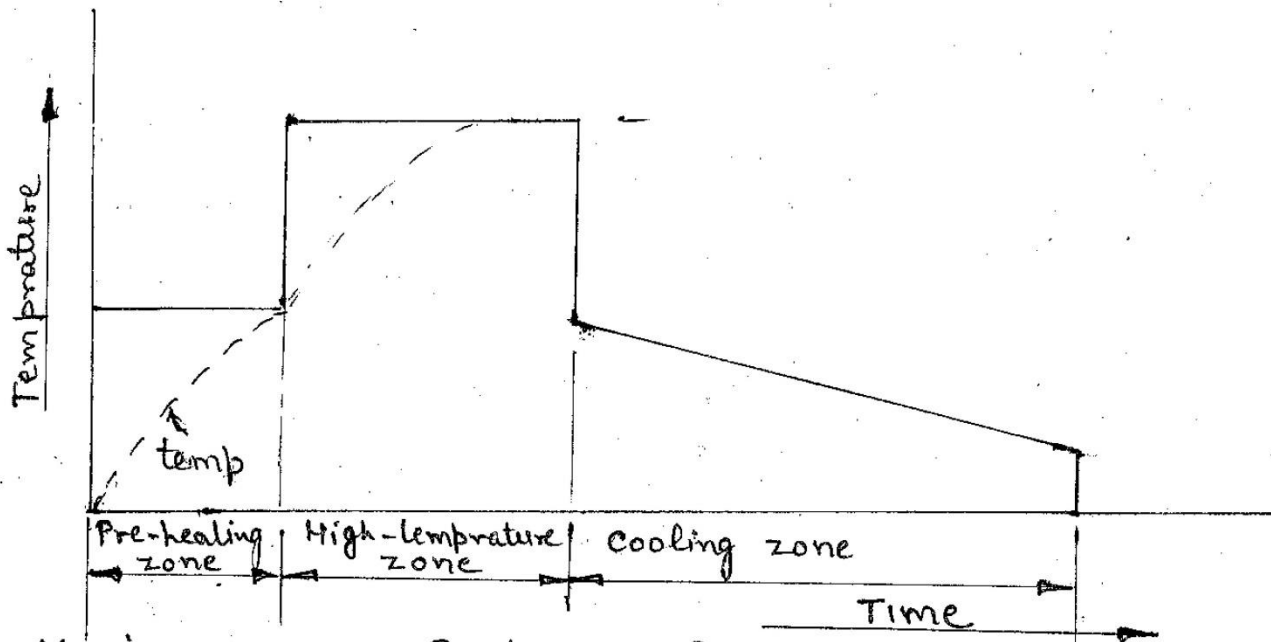
Sintering consists of the following three stages.

1. Pre-heating zone
2. High temperature zone
3. Cooling zone

Pre-heating zone: In this zone the temperature is increased slowly but it should be below sintering temperature. Volatile substances, lubricants and binders are burnt off from the compacted part in this zone.

High temperature Zone: In this zone the temperature of compacted part is raised upto sintering temperature the part is held at this temperature for sufficient time to enable solid state diffusion and bonding between the particles.

Cooling zone: The sintered part is gradually cooled down in the controlled atmosphere of the furnace.



Various zones of Sintering Process in continuous furnaces.

objectives of Sintering

- To achieve high strength
- To achieve good bonding of powdered particles
- To provide a dense and compacted structure of the part
- To produce parts free from oxides
- To increase mechanical, electrical and physical properties of the part.

Secondary Processes.

Most of the Products obtained after sintering can be used as usual but where very close tolerances and better surface finishing is required secondary processes are used. which are given below.

1. Sizing: In this operation the sintered part is placed in a die of accurate dimensions and repressed to bring it to accurate size.
2. Coining: In this operation the sintered part is also repressed in a die, to close voids and to achieve the desired density.
3. Machining: This operation is used to produce some extra features like small holes, threads grooves and undercuts or in order to hold very close tolerances, which are not possible through primary processes.
4. Infiltration: In this operation a sheet of brass or copper having the shape of compacted part is placed upon the pre-sintered part and reheated. The sheet melts and the molten metal fills the pores of the part by capillary action. It increases the density, strength and hardness which is not possible through sintering alone.
5. Impregnation: In this operation the ^{parts} are placed in a tank containing lubricant and heated about $90^{\circ}\text{C} - 100^{\circ}\text{C}$ and held at this temperature about 20 minutes. The lubricant enters into the pores of the part through capillary action imparts the self-lubricating properties to the parts.

6. Heat treatment: In this operation the structures of sintered ferrous parts are improved with the help of heat treatment processes which increases the mechanical properties of the parts like strength, hardness, wear resistance and grain structure etc.
7. Plating: In this operation plating of sintered part is done to prevent corrosion and provide protective atmosphere.

Properties or characteristics of metal Powders.

The main characteristics of metal powders which make them suitable for the process are listed below:

1. Particles size and distribution: Particle size and distribution play very important role over porosity, density, compressibility and shrinkage. If size of particles is same then porosity of parts increases otherwise decreases. If particles have fine grains density of the part increases otherwise decreases.
2. Flow rate: It is the time taken by a definite quantity of powder to flow through a standard orifice. There are two main parameters that are governed by flow rate one is the production rate and other is time required to fill the mould. It depends upon size, shape and distribution of particles.
3. Compressibility: It is defined as the ratio of initial volume to the final volume of the part after compaction. It imparts the green strength which is known as mechanical strength of compact part.

4 Sintering ability: It is the ability of metal powders which imparts the adhesion between them. It should be high enough to produce good quality products. Regular shapes have good sintering properties.

5 Purity : Powders must be pure. Impurities present in powders affects sintering and Compacting operations which results lowering the mechanical properties of the part and also reduces the life of dies.

6. Particle Shape: The shape of particles depends upon the method of production. It may be flat irregular, angular, dendritic, spherical etc. The irregular shaped particles achieve good green strength. The shape of particles affects compactness, strength and porosity.

7. Apparent density: It is the specific gravity of the powder. It depends upon various factors like size, shape, chemical composition and distribution of particles. A portion of component can be made hard and dense by controlling this property.

Advantages of Powder Metallurgy.

1. The wastage of material is negligible.
2. High production rate.
3. Not so much skilled labour is required.
4. Mixtures of metals and non-metals or two immiscible metals can be easily produced to achieve desired properties.
5. Quite complex shapes can be produced.
6. Impossible parts like super-hard cutting tool bits can be produced which can never be produced by any other process.
7. Dimensional accuracy and surface finishing obtained are such that for many applications all machining operations can be eliminated.
8. The products of powder metallurgy are highly pure.
9. Wide range of components with special electrical and magnetic properties can be produced.
10. Parts such as self-lubricating bearings and sintered carbides can be produced by P/M process which can not be produced by any other process.
11. The use of diamond in industry has been made possible mainly through powder metallurgy process.
12. Structure and properties of products can be controlled more closely than in other fabricating processes.

Disadvantages and limitations of Powder Metallurgy.

1. Process is expensive due to high cost of tools & dies.
2. The size of products is limited as compared to casting.
3. Powder metallurgy is not economical for small scale production.
4. It may be difficult some times to obtain particular alloy powders.
5. Parts pressed from the top tend to be less dense at the bottom.
6. Extreme care is required in handling pyrophoric powders (like Mg, Th, Zr) to prevent fires or explosions and toxic powder like U, Be, Th) to minimise health hazard.
7. Parts made by P/M process in most cases do not have as good physical properties as wrought or cast parts.
8. Powder metallurgy is not suitable for a few products.

Application of Powder Metallurgy

Powder metallurgy process is used to produce the following products.

1. Porous products. Such as self lubricating bearing and filters are made by this process.
2. Refractory Parts: The parts made out of tungsten, tantalum and molybdenum are used in electric bulbs, radio valves, oscillator valves, x-ray tubes in the form of filaments, cathode, anode and control grids etc.
3. Special carbide tools: Carbides of tungsten and molybdenum are used for making gauges, wire drawing dies, deep drawing dies, wire guides, stamping and blanking tools, rock drill bits and stone hammers etc.
4. Diamond tools: These tools are made up of mixture of diamond and iron powders such as tool used for cutting glass.
Grinding wheels that are incorporate steel and diamond powder may be produce by powder metallurgy
5. Tungsten parts, ^{made by this process} are also used in plasma jet engines which are operated at high temperature about 1850°C . Silver infiltrated tungsten is used in nozzles for rockets and missiles.
6. Parts made by powder metallurgy process are also used in clock and timing devices, type writers, calculators, permanent magnets, laminated bimetallic strips etc.
7. Products where the combined properties of two metals or of a metal and non-metal are required such as non-porous bearings, electric motor brushes.

8 Electric Contact Parts: These are the parts having high current capacity, low contact resistance and high thermal conductivity, along with high compressive strength and hardness, which are used for circuit cut off, relays and resistance welding electrodes are also made by this process.

9 Atomic energy applications